

Free Fall: Velocity and Displacement

1. Free Fall from Rest: Starting at $t = 0$

Choose UP as positive. On Earth, free-fall acceleration is $a = -10 \text{ m/s}^2$.

1. Earth $a = -10 \text{ m/s}^2$ $v_0 = 0 \text{ m/s}$ $t_1 = 0 \text{ s}$ $t_2 = 1 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

2. Earth $a = -10 \text{ m/s}^2$ $v_0 = 0 \text{ m/s}$ $t_1 = 0 \text{ s}$ $t_2 = 2 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

3. Earth $a = -10 \text{ m/s}^2$ $v_0 = 0 \text{ m/s}$ $t_1 = 0 \text{ s}$ $t_2 = 3 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

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2. Free Fall from Rest: Time Intervals

UP is positive. The object was released from rest at $t = 0$, but your interval may begin later.

4. Earth

$$a = -10 \text{ m/s}^2$$

$$v_0 = 0 \text{ m/s}$$

$$t_1 = 1 \text{ s}$$

$$t_2 = 2 \text{ s}$$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

5. Earth

$$a = -10 \text{ m/s}^2$$

$$v_0 = 0 \text{ m/s}$$

$$t_1 = 1 \text{ s}$$

$$t_2 = 3 \text{ s}$$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

6. Low-gravity planet

$$a = -5 \text{ m/s}^2$$

$$v_0 = 0 \text{ m/s}$$

$$t_1 = 1 \text{ s}$$

$$t_2 = 3 \text{ s}$$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

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3. Free Fall from Rest: More Practice

UP is positive. Negative velocity and negative displacement both indicate downward motion.

7. Earth

$$a = -10 \text{ m/s}^2$$

$$v_0 = 0 \text{ m/s}$$

$$t_1 = 0.5 \text{ s}$$

$$t_2 = 1.5 \text{ s}$$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

8. Earth

$$a = -10 \text{ m/s}^2$$

$$v_0 = 0 \text{ m/s}$$

$$t_1 = 1.2 \text{ s}$$

$$t_2 = 2.4 \text{ s}$$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

9. Heavy planet

$$a = -15 \text{ m/s}^2$$

$$v_0 = 0 \text{ m/s}$$

$$t_1 = 0.4 \text{ s}$$

$$t_2 = 1.6 \text{ s}$$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

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4. Initial Velocity Downward

UP is positive, so an object thrown downward begins with a negative initial velocity.

10. Earth $a = -10 \text{ m/s}^2$ $v_0 = -5 \text{ m/s}$ $t_1 = 0 \text{ s}$ $t_2 = 1.5 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

11. Earth $a = -10 \text{ m/s}^2$ $v_0 = -12 \text{ m/s}$ $t_1 = 0.5 \text{ s}$ $t_2 = 1.5 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

12. Alien planet $a = -8 \text{ m/s}^2$ $v_0 = -4 \text{ m/s}$ $t_1 = 1 \text{ s}$ $t_2 = 2.5 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

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5. Mixed: Dropped or Thrown Downward

UP is positive. Read v_0 carefully: some objects start from rest and others are already moving downward.

13. Earth $a = -10 \text{ m/s}^2$ $v_0 = 0 \text{ m/s}$ $t_1 = 0.8 \text{ s}$ $t_2 = 2 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

14. Earth $a = -10 \text{ m/s}^2$ $v_0 = -6 \text{ m/s}$ $t_1 = 0.5 \text{ s}$ $t_2 = 1.8 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

15. Strong-gravity planet $a = -16 \text{ m/s}^2$ $v_0 = -4 \text{ m/s}$ $t_1 = 0.5 \text{ s}$ $t_2 = 1.5 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

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6. Initial Velocity Upward

UP is positive. These objects begin with positive velocity, while gravity still gives a negative acceleration.

16. Earth $a = -10 \text{ m/s}^2$ $v_0 = 30 \text{ m/s}$ $t_1 = 0.5 \text{ s}$ $t_2 = 2 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

17. Earth $a = -10 \text{ m/s}^2$ $v_0 = 40 \text{ m/s}$ $t_1 = 1 \text{ s}$ $t_2 = 3 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

18. Alien planet $a = -8 \text{ m/s}^2$ $v_0 = 24 \text{ m/s}$ $t_1 = 0.5 \text{ s}$ $t_2 = 2 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

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7. Up, Stop, Then Down

UP is positive. The same negative acceleration acts before, at, and after the highest point.

19. Earth $a = -10 \text{ m/s}^2$ $v_0 = 20 \text{ m/s}$ $t_1 = 1 \text{ s}$ $t_2 = 3 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

20. Earth $a = -10 \text{ m/s}^2$ $v_0 = 30 \text{ m/s}$ $t_1 = 2 \text{ s}$ $t_2 = 4 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

21. Alien planet $a = -6 \text{ m/s}^2$ $v_0 = 18 \text{ m/s}$ $t_1 = 2 \text{ s}$ $t_2 = 4 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

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8. Putting It All Together

UP is positive. Use the signs of velocity, average velocity, and displacement to describe what happened.

22. Earth $a = -10 \text{ m/s}^2$ $v_0 = 20 \text{ m/s}$ $t_1 = 0.5 \text{ s}$ $t_2 = 3.5 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

After calculating, use the signs to decide the direction of the net motion during the interval.

23. Earth $a = -10 \text{ m/s}^2$ $v_0 = -8 \text{ m/s}$ $t_1 = 0.8 \text{ s}$ $t_2 = 2.2 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

After calculating, use the signs to decide the direction of the net motion during the interval.

24. Alien planet $a = -8 \text{ m/s}^2$ $v_0 = 24 \text{ m/s}$ $t_1 = 1 \text{ s}$ $t_2 = 5 \text{ s}$

Find: v_1 , v_2 , average velocity from t_1 to t_2 , and displacement during that interval.

After calculating, use the signs to decide the direction of the net motion during the interval.